Topic 1.7 - Rational Functions and End Behavior

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: ____________________ Class: ____________ Date: ____________

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Determine rational end behavior by comparing numerator and denominator degrees.
- Use polynomial division to identify slant or polynomial asymptotes.
- Analyze approach from above or below and interpret long-run behavior in context.

Section A: Foundation and representation

Question 1

Foundation | No calculator

Find the horizontal asymptote of $\frac{5x^3-2x+1}{-2x^3+7}$.

Question 2

Degree comparison | No calculator

Find the horizontal asymptote of $\frac{4x^2+1}{x^5-3}$.

Section A continued

Question 3

Reasoning No calculator

Explain why $\frac{x^4+1}{2x^2-5}$ does not have a horizontal or slant asymptote.

Question 4

Division No calculator

Divide $\frac{3x^2-5x+7}{x+1}$ and state the slant asymptote.

Question 5

Division No calculator

Find the polynomial asymptote of $\frac{x^4+2x^3-x+1}{x^2+1}$.

Section B: Application and reasoning

Question 6

Approach | No calculator

For $R = 4 - \frac{7}{x^2+2}$, describe approach to the asymptote.

Question 7

Approach | No calculator

For $S = -3 + \frac{x}{x^2+9}$, describe approach on each end.

Question 8

Construction | No calculator

Construct a rational function with horizontal asymptote $y = 6$, no real vertical asymptotes, and $R(0) = 4$.

Section C: Challenge and error analysis

Question 9

Error analysis | No calculator

A student says $\frac{2x^3+1}{x^2+5}$ has horizontal asymptote $y = 2$ because of leading coefficients. Diagnose.

Question 10

Parameter | No calculator

Find k so $\frac{kx^2+3x}{2x^2-1}$ has horizontal asymptote $y = -4$.

Question 11

Context | No calculator

Average cost is $C(n) = \frac{1200+8n}{n+15}$. Find its long-run value and interpret.

Question 12

Challenge | No calculator

Construct a rational function with asymptote $y = x^2 - 3$, lying above it for all real x in its domain and having no real exclusions.

Section D: AP-style multiple choice

MCQ 1

No calculator

What is the horizontal asymptote of $\frac{-6x^4+x}{3x^4+2}$?

A. $y = -6$

B. $y = -2$

C. $y = 0$

D. No horizontal asymptote

MCQ 2

No calculator

Which function has slant asymptote $y = 2x + 1$?

A. $2x + 1 + \frac{3}{x^2+1}$

B. $\frac{2x+1}{x^2+1}$

C. $2 + \frac{x}{x^2+1}$

D. $\frac{x^2}{2x+1}$

MCQ 3

No calculator

If numerator degree is two less than denominator degree, the horizontal asymptote is

A. the leading ratio

B. $y = 0$

C. a slant line

D. a quadratic

MCQ 4

No calculator

For $R = 5 + \frac{-3x}{x^2+1}$, how does R approach 5 as $x \to \infty$?

A. from above

B. from below

C. does not approach

D. alternates

AP-style free response 1

No calculator - 6 points

Let $R(x) = \frac{2x^3-x^2+5}{x^2-4}$.

A. Write quotient-remainder form.

B. State the polynomial asymptote and domain exclusions.

C. Determine approach relative to the asymptote as $x \to \infty$.

D. Determine approach relative to the asymptote as $x \to -\infty$.

AP-style free response 2

Calculator permitted - 6 points

Average cost, in QAR per item, is $C(n) = \frac{1200+8n}{n+15}$ for positive integer production n.

A. Find and interpret the horizontal asymptote.

B. Calculate $C(100)$ to three decimals.

C. Solve $C(n) = 12$ and interpret the solution.

D. Explain why the model should not be evaluated at $n = -15$.